Scenario For The Descriptive,Diagnostic,Predictive and Prescriptive Analysis

1. Industrial Gas Turbine

- **Sensor Values:**
    - Exhaust temperature spread: **52°C** (Normal is below 25°C)
    - Power output: **95.8 MW** (Expected is 100 MW)
- **Descriptive (*What happened?*):**
    - The dashboard shows that the turbine's exhaust temperature became uneven across different points, and total electrical power dropped by 4.2% during peak operation.
- **Diagnostic (*Why did it happen?*):**
    - By comparing fuel flow and pipe pressure readings, the analysis found that fuel nozzle #3 was clogged with carbon deposits, starving that specific burner of fuel.
- **Predictive (*What will happen next?*):**
    - Trend forecasting models predict that uneven heat stress will cause a micro-crack in turbine blade #12 within **380 operating hours** if left unaddressed.
- **Prescriptive (*What should we do?*):**
    - Automatically reduce the turbine load by 6% to keep operating temperatures safe, and schedule a technician to clean the nozzle on Tuesday at 2:00 AM when electricity demand is lowest.

2. Smart Elevator

- **Sensor Values:**
    - Door motor current draw: **4.2 Amps** (Normal baseline is 2.8 Amps)
    - Door closing duration: **4.8 seconds** (Normal is 2.5 seconds)
- **Descriptive (*What happened?*):**
    - Event logs show that the elevator doors on the 14th floor took nearly twice as long to close, causing morning passenger wait times to jump to 3.5 minutes.
- **Diagnostic (*Why did it happen?*):**
    - Looking at electrical current curves, the motor spiked in power usage at the 70% closed point, indicating physical resistance caused by dirt and debris stuck in the bottom floor track.
- **Predictive (*What will happen next?*):**
    - The model projects that the struggling motor will overheat and lock up completely within **65 door cycles** (around 2 hours of normal traffic).

- **Prescriptive (*What should we do?*):**
    - Send the elevator to park on floor 14 with doors closed, reroute passenger calls to the remaining elevators, and automatically send a cleanup alert to the building technician's phone.

3. Automated Greenhouse

- **Sensor Values:**
    - Soil moisture level: **16% saturation** (Target range is 40%–50%)
    - Pipe water flow meter: **0.0 Liters/min** (Valve status is set to OPEN)
- **Descriptive (*What happened?*):**
    - Sensor charts show that soil moisture in Zone 3 dried out rapidly over 6 hours, while plant leaf temperatures rose by 2.8°C.
- **Diagnostic (*Why did it happen?*):**
    - Comparing the open valve command with zero water flow down the line revealed that mineral buildup had completely blocked the main water filter.
- **Predictive (*What will happen next?*):**
    - Growth and soil moisture models predict the plants will reach a permanent wilting point within **4.5 hours**, leading to a 30% crop loss.
- **Prescriptive (*What should we do?*):**
    - Turn on overhead misting nozzles for 20 seconds every 15 minutes to keep plants hydrated, dim grow lights by 35% to reduce heat stress, and switch water through a backup pipeline.

4. Hospital MRI Scanner

- **Sensor Values:**
    - Liquid helium boil-off rate: **0.45 Liters/day** (Normal is 0.05 Liters/day)
    - Compressor cold-head pressure: **18.8 bar** (Normal limit is 17.5 bar)
- **Descriptive (*What happened?*):**
    - Maintenance charts show the MRI magnet is losing liquid helium 9 times faster than usual, and internal cooling pressure is rising.
- **Diagnostic (*Why did it happen?*):**
    - Vibration and sound analysis of the cooling compressor detected abnormal mechanical noise, pointing to a worn-out piston seal that can no longer turn helium gas back into liquid.

- **Predictive (*What will happen next?*):**
    - Thermodynamic pressure models show that helium levels will reach an emergency safety shutoff (quench) within **6 days**, risking an expensive emergency system shutdown.
- **Prescriptive (*What should we do?*):**
    - Limit long brain scan sessions to 30 minutes to reduce heat generation, order a replacement piston seal kit, and schedule maintenance for Saturday evening when no patients are booked.

5. Water Desalination Plant

- **Sensor Values:**
    - Membrane pressure difference (ΔP): **2.3 bar** (Normal is 1.6 bar)
    - Clean freshwater flow: **88 m^3/hour** (Target baseline is 100 m^3/hour)
- **Descriptive (*What happened?*):**
    - Daily production logs show that the pressure needed to push water through the filtration membranes jumped by 0.7 bar, while clean water output dropped by 12%.
- **Diagnostic (*Why did it happen?*):**
    - Cross-checking chemical dosing logs against water mineral levels revealed that the anti-scaling chemical pump ran 15% too low over the weekend, allowing salt minerals to crust over the filter sheets.
- **Predictive (*What will happen next?*):**
    - Membrane decay models forecast that the mineral buildup will become permanently baked into the filter within **72 hours**, requiring full replacement of the $15,000 membrane filters.
- **Prescriptive (*What should we do?*):**
    - Divert incoming seawater to backup filter Bank B, isolate the dirty filters, and start an automated chemical wash tonight at 11:00 PM when electricity costs are lowest.

Product	Sensor Readings	Descriptive (What?)	Diagnostic (Why?)	Predictive (What next?)	Prescriptive (What to do?)
Gas Turbine	Exhaust spread: **52°C**; Output: **95.8 MW**	Heat is uneven; MW output dropped 4.2%	Clogged nozzle #3 restricting gas flow	Blade cracking within 380 operating hours	Reduce load 6%; clean nozzle at off-peak 2:00 AM
Smart Elevator	Motor current: **4.2A**; Door close: **4.8s**	Doors closing slowly; wait times reached 3.5 min	Dirt in bottom door sill causing motor drag	Motor overheat shutoff within 65 door cycles	Park car at floor 14; reroute calls; alert tech
Greenhouse	Soil moisture: **16%**; Flow: **0.0 L/min**	Soil dried out; leaf temperatures climbed 2.8°C	Mineral buildup blocked main filter line	Permanent plant wilting in 4.5 hours	Run misting; dim lights 35%; open backup line
MRI Scanner	Helium loss: **0.45 L/day**; Pressure: **18.8 bar**	Helium evaporating fast; pressure rising	Worn compressor piston seal not condensing gas	Emergency safety quench within 6 days	Shorten scans; stage replacement seal for Saturday
Desalination Plant	Membrane ΔP: **2.3 bar**; Flow: **88 m³/h**	Water output dropped 12%; pressure climbed	Low anti-scale chemical dose caused salt crusting	Permanent filter damage within 72 hours	Switch to Bank B; run chemical wash at 11:00 PM